

08-07-2026

Agenda:

ML-16

→ Naive Bayes,
 ↗ categorical column (practical) •
 ↘ numerical column

→ Kmean clustering (Theory)
 → practical

cv → accuracy → Train → 50%.

97% 96% 98%

test → 100% →

→ 100% 96% 99%

91% 92% 89% 94%

100% 100% 100 100

K-means Clustering (Unsupervised learning)

↓
no target column

$X \rightarrow Y$
 ↗ regression → evaluation metrics
 ↘ classification
 ↓
 will not work

Data → K-means - Clustering → groups based on similarity (clustering)

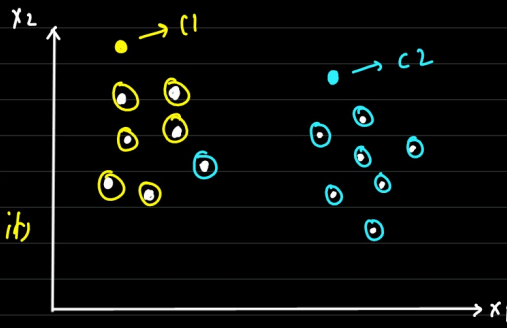
↑
how many groups
you want?

↑
K

Partition:

$k=2$

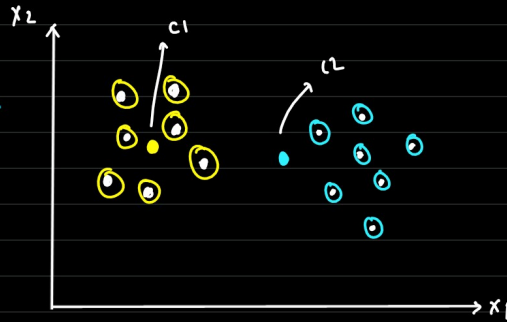
- randomly init centroid
- select a point & check its distance from centroid
- assign the point to the closest centroid
- do for all data point
- compute new location for centroid



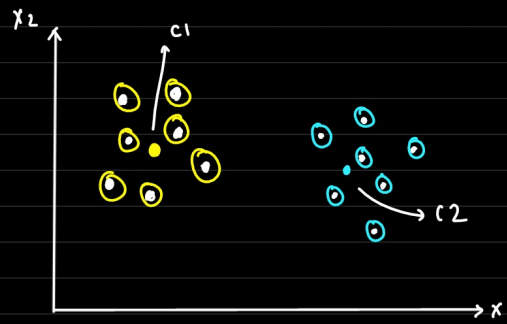
→ iteration - 1

when should we stop training?

→ when no new data points are being assigned to a cluster.



itr-1	itr-2	i3	i4
c1 → 5	c1 → 6	6	6
c2 → 6	c1 → 5	5	5



Data set :-

$A = (1, 2)$
 $B = (1.5, 1.8)$
 $C = (5, 8)$
 $D = (8, 8)$
 $E = (1, 0.6)$
 $F = (9, 11)$

iteration - 1

(1) point - A $(1, 2)$

$$\text{to } c1: \sqrt{(1-1)^2 + (2-2)^2} = 0$$

$(1, 2)$

$$\text{to } c2: \sqrt{(1-5)^2 + (2-8)^2} = 7.211$$

$(5, 8)$

→ assign A → cluster C1

(2) point B $(1.5, 1.8)$

$$\text{to } c1: \sqrt{(1.5-1)^2 + (1.8-2)^2} \approx 0.5385$$

$$\text{to } c2: \sqrt{(1.5-5)^2 + (1.8-8)^2} \approx 7.119$$

→ assign B → cluster C1

$c1 = \{A, B\}$

Hyper-parameter

$k=2$

$c1 \rightarrow A = (1, 2)$

$c2 \rightarrow C = (5, 8)$

method-1 $\rightarrow$ pick random
data point as
the centroid coordinate

(3) point C $\rightarrow$ c2
(4) point D $\rightarrow$ c2
(5) point E $\rightarrow$ c1
(6) point F $\rightarrow$ c2

step 2: update centroids

New c1: mean of A (1, 2)
B (1.5, 1.8)
E (1, 0.6)

$$c1_{\text{new}} = \left(\frac{1+1.5+1}{3}, \frac{2+1.8+0.6}{3} \right)$$

$\downarrow$
1.667, 1.4667

method-2 $\rightarrow$ kmeans++

k=2

c1 $\rightarrow$ random $\rightarrow$ step-1

c2 $\rightarrow$ (c1, all data point) $\rightarrow$ step-2

c1 = [A, B, E]
c2 = [C, D, F]

Euclidean Distance

(x1, x2) (A1, A2)

$$\sqrt{(x_1 - A_1)^2 + (x_2 - A_2)^2}$$

c1_old = (1, 2)
c1_new = (1.667, 1.4667)

c2_old = (5, 8)
c2_new = (7.33, 9)

iteration-2:

c1_new = (1.667, 1.4667)

c2_new = (7.33, 9)

iteration-2

(1) point - A (1, 2)

to c1:

to c2:

c1 = [A, B, E] $\leftarrow$ final c1 & c2
c2 = [C, D, F]

k=2

k=1 $\rightarrow$ all data $\rightarrow$ single group

k = Total No of data points

$\downarrow$

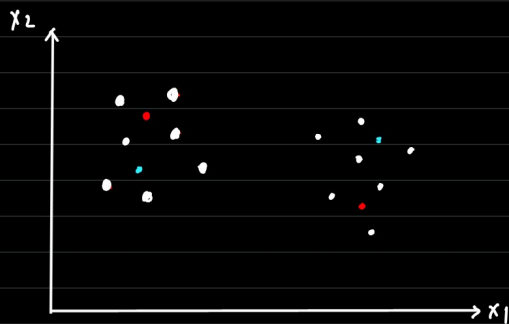
each point will be assigned to a cluster,

6 point $\rightarrow$ k=6

k=N $\rightarrow$ each data $\rightarrow$ centroid

optimal value of k:

WCSS $\rightarrow$ within cluster summation square



$k=1 \rightarrow$ sum all the distance from centroid
 $\approx$ high



$k=N$, where $N = \text{no of data point}$

$\approx$ close to 0

$\approx$ low value

$k=1 \rightarrow w(1) \rightarrow \square$

$k=2 \rightarrow w(1) \rightarrow \begin{matrix} \square \\ \downarrow \\ \square \end{matrix} \rightarrow \square$

$k=3 \rightarrow w(1) \rightarrow \begin{matrix} \square \\ \square \\ \square \end{matrix} \rightarrow \square$

$w(1)$: —

$c_1 \rightarrow A, B, E$

$$(A-c_1)^2 + (B-c_1)^2 + (E-c_1)^2$$

≈ 1.3163

$c_2 \rightarrow C, D, F$

$$(C-c_2)^2 + (D-c_2)^2 + (F-c_2)^2$$

14.666

$k=2 \rightarrow \begin{matrix} \square \\ + \\ \square \end{matrix}$

$$\begin{matrix} \swarrow & \nwarrow \\ w(1)(c_1) & + & w(1)(c_2) & \leftarrow & w(1)(k=2) \\ & (15.982) \end{matrix}$$

$k=1$
 $k=2$
 $k=3$
 $k=4$

$w(1)$
 $\square$
 $\square$
 $\square$
 $\square$

} ?